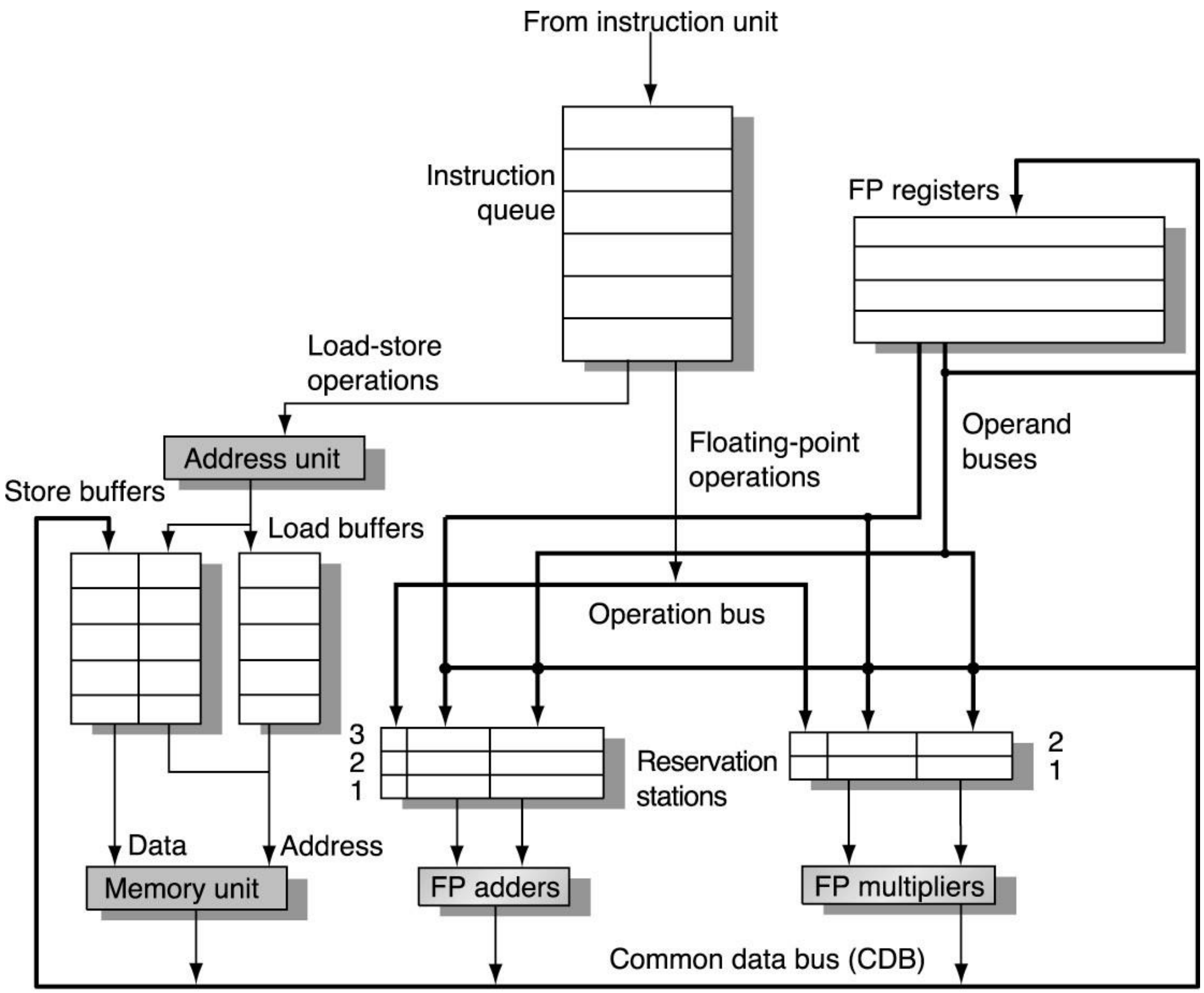
Examples of behavior of the Tomasulo’s architecture

Tomasulo Organization



Reservation Station Components

Op—Operation to perform in the unit (e.g., + or –)

Qj, Qk—Reservation stations producing source registers

Vj, Vk—Value of Source operands

Rj, Rk—Flags indicating when Vj, Vk are ready

Busy—Indicates reservation station and FU is busy

Register result status—Indicates which functional unit will write each register, if one exists. Blank when no pending instructions that will write that register.

Three Stages of Tomasulo Algorithm

1. **Issue—get instruction from FP Op Queue**

   If reservation station free, the scoreboard issues instr & sends operands (renames registers).

2. **Execution—operate on operands (EX)**

   When both operands ready then execute; if not ready, watch CDB for result

3. **Write result—finish execution (WB)**

   Write on Common Data Bus to all awaiting units; mark reservation station available.

Tomasulo Example Cycle 0

Instruction Status

Instruction		j	k	Issue	Execution Complete	Write Result
LD	**F6**	**34+**	**R2**			
LD	**F2**	**45+**	**R3**			
MULTD	**F0**	**F2**	**F4**			
SUBD	**F8**	**F6**	**F2**			
DIVD	**F10**	**F0**	**F6**			
ADDD	**F6**	**F8**	**F2**			

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Resevation Stations

Time	Name	Busy	OP	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
0	Mult1	**No**					
0	Mult2	**No**					

Register result status

Clock		F0	F2	F4	F6	F8	F10	F12	…	F30
0	FU									

Tomasulo Example Cycle 1

Instruction Status					Execution	Write				
Instruction		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1				Load1	Yes	34+R2
LD	**F2**	**45+**	**R3**					Load2	No	
MULTD	**F0**	**F2**	**F4**					Load3	No	
SUBD	**F8**	**F6**	**F2**							
DIVD	**F10**	**F0**	**F6**							
ADDD	**F6**	**F8**	**F2**							

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
0	Mult1	**No**					
0	Mult2	**No**					

Register result status		F0	F2	F4	F6	F8	F10	F12	…	F30
Clock										
1	FU				Load1					

Tomasulo Example Cycle 2

Instruction Status					Execution	Write			
Instruction		j	k	Issue	Complete	Result		Busy	Address
LD	**F6**	**34+**	**R2**	1			Load1	Yes	34+R2
LD	**F2**	**45+**	**R3**	2			Load2	Yes	45+R3
MULTD	**F0**	**F2**	**F4**				Load3	No	
SUBD	**F8**	**F6**	**F2**						
DIVD	**F10**	**F0**	**F6**						
ADDD	**F6**	**F8**	**F2**						

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
0	Mult1	**No**					
0	Mult2	**No**					

Register result status

Clock		F0	F2	F4	F6	F8	F10	F12	...	F30
2	FU		Load2		Load1					

Tomasulo Example Cycle 3

Instruction Status					Execution	Write				
Instruction		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3			Load1	Yes	34+R2
LD	**F2**	**45+**	**R3**	2				Load2	Yes	45+R3
MULTD	**F0**	**F2**	**F4**	3				Load3	No	
SUBD	**F8**	**F6**	**F2**							
DIVD	**F10**	**F0**	**F6**							
ADDD	**F6**	**F8**	**F2**							

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	0	Add2	**No**					
	0	Add3	**No**					
	0	Mult1	**Yes**	MULTD		R[F4]	Load2	
	0	Mult2	**No**					

Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	…	F30
3			FU	Mult1	Load2		Load1					

Tomasulo Example Cycle 4

Instruction Status													
					Execution	Write							
Instruction		j	k	Issue	Complete	Result			Busy	Address			
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No				
LD	**F2**	**45+**	**R3**	2	4			Load2	Yes	45+R3			
MULTD	**F0**	**F2**	**F4**	3				Load3	No				
SUBD	**F8**	**F6**	**F2**	4									
DIVD	**F10**	**F0**	**F6**										
ADDD	**F6**	**F8**	**F2**										

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**Yes**	SUBD	M[34+R2]			Load2
	0	Add2	**No**					
	0	Add3	**No**					
	0	Mult1	**Yes**	MULTD		R[F4]	Load2	
	0	Mult2	**No**					

Register result status													
Clock				F0	F2	F4	F6	F8	F10	F12	…	F30
4			FU	Mult1	Load2		M[34+R2]	Add1				

Tomasulo Example Cycle 5

Instruction Status					Execution	Write				
Instruction		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3				Load3	No	
SUBD	**F8**	**F6**	**F2**	4						
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**							

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**Yes**	SUBD	M[34+R2]	M[45+R3]		
	0	Add2	**No**					
	0	Add3	**No**					
	0	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
	0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
5			FU	Mult1	M[45+R3]		M[34+R2]	Add1	Mult2			

Tomasulo Example Cycle 6

Instruction Status					Execution	Write			
Instruction		j	k	Issue	Complete	Result		Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4	Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5	Load2	No	
MULTD	**F0**	**F2**	**F4**	3			Load3	No	
SUBD	**F8**	**F6**	**F2**	4					
DIVD	**F10**	**F0**	**F6**	5					
ADDD	**F6**	**F8**	**F2**	6					

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
2	Add1	**Yes**	SUBD	M[34+R2]	M[45+R3]		
0	Add2	**Yes**	ADDD		M[45+R3]	Add1	
0	Add3	**No**					
10	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status		F0	F2	F4	F6	F8	F10	F12	…	F30
Clock										
6	FU	Mult1	M[45+R3]		Add2	Add1	Mult2			

Tomasulo Example Cycle 7

Instruction Status					Execution	Write		Busy	Address
Instruction		j	k	Issue	Complete	Result		Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4	Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5	Load2	No	
MULTD	**F0**	**F2**	**F4**	3			Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7				
DIVD	**F10**	**F0**	**F6**	5					
ADDD	**F6**	**F8**	**F2**	6					

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
1	Add1	**Yes**	SUBD	M[34+R2]	M[45+R3]		
0	Add2	**Yes**	ADDD		M[45+R3]	Add1	
0	Add3	**No**					
9	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status		F0	F2	F4	F6	F8	F10	F12	…	F30
Clock		F0	F2	F4	F6	F8	F10	F12	…	F30
7	FU	Mult1	M[45+R3]		Add2	Add1	Mult2			

Tomasulo Example Cycle 8

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3				Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6						

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	0	Add2	**Yes**	ADDD	M()-M()	M[45+R3]		
	0	Add3	**No**					
	8	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
	0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status											
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
8			FU	Mult1	M[45+R3]		Add2	M()-M()	Mult2			

Tomasulo Example Cycle 9

Instruction Status					Execution	Write			
		j	k	Issue	Complete	Result		Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4	Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5	Load2	No	
MULTD	**F0**	**F2**	**F4**	3			Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8			
DIVD	**F10**	**F0**	**F6**	5					
ADDD	**F6**	**F8**	**F2**	6					

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
0	Add1	**No**					
2	Add2	**Yes**	ADDD	M[]-M[]	M[45+R3]		
0	Add3	**No**					
7	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status									
Clock		F0	F2	F4	F6	F8	F10	F12	…	F30
9	FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 10

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3				Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6	10					

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	1	Add2	**Yes**	ADDD	M[]-M[]	M[45+R3]		
	0	Add3	**No**					
	6	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
	0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status											
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
10			FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 11

Instruction Status		j	k	Issue	Execution Complete	Write Result
LD	**F6**	**34+**	**R2**	1	3	4
LD	**F2**	**45+**	**R3**	2	4	5
MULTD	**F0**	**F2**	**F4**	3		
SUBD	**F8**	**F6**	**F2**	4	7	8
DIVD	**F10**	**F0**	**F6**	5		
ADDD	**F6**	**F8**	**F2**	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Resevation Stations

Time	Name	Busy	OP	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
5	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status

Clock		F0	F2	F4	F6	F8	F10	F12	…	F30
11	FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 12

Instruction Status		j	k	Issue	Execution Complete	Write Result
LD	**F6**	**34+**	**R2**	1	3	4
LD	**F2**	**45+**	**R3**	2	4	5
MULTD	**F0**	**F2**	**F4**	3		
SUBD	**F8**	**F6**	**F2**	4	7	8
DIVD	**F10**	**F0**	**F6**	5		
ADDD	**F6**	**F8**	**F2**	6	10	11

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Resevation Stations

Time	Name	Busy	OP	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
4	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status

Clock		F0	F2	F4	F6	F8	F10	F12	…	F30
12	FU	Mult1	M[45+R3]		Add2	M[]-M[]	Mult2			

Tomasulo Example Cycle 13

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3				Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6	10	11				

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	0	Add2	**No**					
	0	Add3	**No**					
	3	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
	0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status											
Clock				F0	F2	F4	F6	F8	F10	F12	…	F30
13			FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 14

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3				Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6	10	11				

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	0	Add2	**No**					
	0	Add3	**No**					
	2	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
	0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
14			FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 15

Instruction Status		j	k	Issue	Execution Complete	Write Result		Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4	Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5	Load2	No	
MULTD	**F0**	**F2**	**F4**	3	15		Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8			
DIVD	**F10**	**F0**	**F6**	5					
ADDD	**F6**	**F8**	**F2**	6	10	11			

Resevation Stations

Time	Name	Busy	OP	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
1	Mult1	**Yes**	MULTD	M[45+R3]	R[F4]		
0	Mult2	**Yes**	DIVD		M[34+R2]	Mult1	

Register result status

Clock		F0	F2	F4	F6	F8	F10	F12	…	F30
15	FU	Mult1	M[45+R3]		(M-M)+M	M[]-M[]	Mult2			

Tomasulo Example Cycle 16

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3	15	16		Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6	10	11				

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	0	Add2	**No**					
	0	Add3	**No**					
	0	Mult1	**No**					
	0	Mult2	**Yes**	DIVD	M*F4	M[34+R2]		

Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
16			FU	M*F4	M[45+R3]		(M-M)+M	M()-M()	Mult2			

Tomasulo Example Cycle 17

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3	15	16		Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6	10	11				

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	**No**					
	0	Add2	**No**					
	0	Add3	**No**					
	0	Mult1	**No**					
	40	Mult2	**Yes**	DIVD	M*F4	M[34+R2]		

Register result status											
Clock				F0	F2	F4	F6	F8	F10	F12	...	F30
17			FU	M*F4	M[45+R3]		(M-M)+M	M()-M()	Mult2			

Tomasulo Example Cycle 18

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4		Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5		Load2	No	
MULTD	**F0**	**F2**	**F4**	3	15	16		Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8				
DIVD	**F10**	**F0**	**F6**	5						
ADDD	**F6**	**F8**	**F2**	6	10	11				

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
0	Mult1	**No**					
39	Mult2	**Yes**	DIVD	M*F4	M[34+R2]		

Register result status											
Clock			F0	F2	F4	F6	F8	F10	F12	…	F30
18		FU	M*F4	M[45+R3]		(M-M)+M	M()-M()	Mult2			

Tomasulo Example Cycle 56

Instruction Status					Execution	Write			
		j	k	Issue	Complete	Result		Busy	Address
LD	**F6**	**34+**	**R2**	1	3	4	Load1	No	
LD	**F2**	**45+**	**R3**	2	4	5	Load2	No	
MULTD	**F0**	**F2**	**F4**	3	15	16	Load3	No	
SUBD	**F8**	**F6**	**F2**	4	7	8			
DIVD	**F10**	**F0**	**F6**	5	56				
ADDD	**F6**	**F8**	**F2**	6	10	11			

Resevation Stations				S1	S2	RS for j	RS for k
Time	Name	Busy	OP	Vj	Vk	Qj	Qk
0	Add1	**No**					
0	Add2	**No**					
0	Add3	**No**					
0	Mult1	**No**					
1	Mult2	**Yes**	DIVD	M*F4	M[34+R2]		

Register result status										
Clock		F0	F2	F4	F6	F8	F10	F12	...	F30
56	FU	M*F4	M[45+R3]		(M-M)+M	M()-M()	Mult2			

Tomasulo Example Cycle 57

Instruction Status					Execution	Write				
		j	k	Issue	Complete	Result			Busy	Address
LD	F6	34+	R2	1	3	4		Load1	No	
LD	F2	45+	R3	2	4	5		Load2	No	
MULTD	F0	F2	F4	3	15	16		Load3	No	
SUBD	F8	F6	F2	4	7	8				
DIVD	F10	F0	F6	5	56	57				
ADDD	F6	F8	F2	6	10	11				

Resevation Stations					S1	S2	RS for j	RS for k
	Time	Name	Busy	OP	Vj	Vk	Qj	Qk
	0	Add1	No					
	0	Add2	No					
	0	Add3	No					
	0	Mult1	No					
	0	Mult2	No					

Register result status												
Clock				F0	F2	F4	F6	F8	F10	F12	…	F30
57			FU	M*F4	M[45+R3]		(M-M)+M	M[]-M[]	M*F4/M			

Tomasulo Loop Example

Loop:	LD	F0	0	R1
	MULTD	F4	F0	F2
	SD	F4	0	R1
	SUBI	R1	R1	#8
	BNEZ	R1	Loop	

- **Multiply takes 4 clock cycles**
- **Loads have cache misses**

Loop Example Cycle 0

Instruction status

Instruction	j	k	iteration	Issue	Execution complete	Write Result		Busy	Address
LD F0	0	R1	1				Load1	No	
MULT F4	F0	F2	1				Load2	No	
SD F4	0	R1	1				Load3	No	Qi
LD F0	0	R1	2				Store1	No	
MULT F4	F0	F2	2				Store2	No	
SD F4	0	R1	2				Store3	No	

Reservation Stations

Time	Name	Busy	Op	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	No					
0	Add2	No					
0	Add3	No					
0	Mult1	No					
0	Mult2	No					

Code:

LD	F0	0	R1
MULT	F4	F0	F2
SD	F4	0	R1
SUBI	R1	R1	#8
BNEZ	R1	Loop	

Register result status

Clock	R1		F0	F2	F4	F6	F8	F10	F12 ...	F30
0	**80**	*Qi*								

Loop Example Cycle 1

Instruction status

Instruction	j	k	iteration	Issue	Execution complete	Write Result		Busy	Address	Qi
LD F0	0	R1	1	1			Load1	Yes	80	
MULT F4	F0	F2	1				Load2	No		
SD F4	0	R1	1				Load3	No		
LD F0	0	R1	2				Store1	No		
MULT F4	F0	F2	2				Store2	No		
SD F4	0	R1	2				Store3	No		

Reservation Stations

Time	Name	Busy	Op	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk	Code:
0	Add1	No						LD F0 0 R1
0	Add2	No						MULT F4 F0 F2
0	Add3	No						SD F4 0 R1
0	Mult1	No						SUBI R1 R1 #8
0	Mult2	No						BNEZ R1 Loop

Register result status

Clock	R1		F0	F2	F4	F6	F8	F10	F12	...	F30
1	**80**	*Qi*	Load1								

Loop Example Cycle 2

Instruction status

Instruction	j	k	iteration	Issue	Execution complete	Write Result
LD F0	0	R1	1	1		
MULT F4	F0	F2	1	2		
SD F4	0	R1	1			
LD F0	0	R1	2			
MULT F4	F0	F2	2			
SD F4	0	R1	2			

	Busy	Address
Load1	Yes	80
Load2	No	
Load3	No	

	Busy	Qi
Store1	No	
Store2	No	
Store3	No	

Reservation Stations

Time	Name	Busy	Op	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	No					
0	Add2	No					
0	Add3	No					
0	Mult1	Yes	MULTD		R(F2)	Load1	
0	Mult2	No					

Code:

LD	F0	0	R1
MULT	F4	F0	F2
SD	F4	0	R1
SUBI	R1	R1	#8
BNEZ	R1	Loop	

Register result status

Clock	R1		F0	F2	F4	F6	F8	F10	F12 ...	F30
2	**80**	Qi	Load1		Mult1					

Loop Example Cycle 3

Instruction status

Instruction	j	k	iteration	Issue	Execution complete	Write Result
LD F0	0	R1	1	1		
MULT F4	F0	F2	1	2		
SD F4	0	R1	1	3		
LD F0	0	R1	2			
MULT F4	F0	F2	2			
SD F4	0	R1	2			

	Busy	Address	Qi
Load1	Yes	80	
Load2	No		
Load3	No		
Store1	Yes	80	Mult1
Store2	No		
Store3	No		

Reservation Stations

Time	Name	Busy	Op	S1 Vj	S2 Vk	RS for j Qj	RS for k Qk
0	Add1	No					
0	Add2	No					
0	Add3	No					
0	Mult1	Yes	MULTD		R(F2)	Load1	
0	Mult2	No					

Code:

```
LD   F0  0  R1
MULT F4  F0 F2
SD   F4  0  R1
SUBI R1  R1 #8
BNEZ R1  Loop
```

Register result status

Clock	R1		F0	F2	F4	F6	F8	F10	F12 ...	F30
3	**80**	Qi	Load1		Mult1					

Loop Example Cycle 4

Instruction status						Executi	Write					
Instructior		j	k	iteration	Issue	complet	Result		Busy	Address		
LD	F0	0	R1	1	1			Load1	Yes	80		
MUL	F4	F0	F2	1	2			Load2	No			
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2				Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2				Store2	No			
SD	F4	0	R1	2				Store3	No			
Reservation Stations					S1	S2	RS for	RS for k				
	Time	Name	Busy	Op	Vj	Vk	Qj	Qk	Code:			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUBI	R1	R1	#8
	0	Mult2	No						BNE	R1	Loop	
Register result status												
Clock		R1		F0	F2	F4	F6	F8	F10	F12	...	F30
4		80	Qi	Load1		Mult1						

Loop Example Cycle 5

Instruction status						Executi	Write					
Instructior		j	k	iteration	Issue	complet	Result		Busy	Address		
LD	F0	0	R1	1	1			Load1	Yes	80		
MUL	F4	F0	F2	1	2			Load2	No			
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2				Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2				Store2	No			
SD	F4	0	R1	2				Store3	No			
Reservation Stations					S1	S2	RS fo	RS for k				
	Time	Name	Busy	Op	Vj	Vk	Qj	Qk	Code:			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUB	R1	R1	#8
	0	Mult2	No						BNE	R1	Loop	
Register result status												
Clock		**R1**		F0	F2	F4	F6	F8	F1C	F1	...	F3C
5		**80**	Qi	Load1		Mult1						

Loop Example Cycle 6

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1			Load1	Yes	80		
MUL	F4	F0	F2	1	2			Load2	Yes	72		
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2	6			Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2				Store2	No			
SD	F4	0	R1	2				Store3	No			
<u>Reservation Stations</u>					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUBI	R1	R1	#8
	0	Mult2	No						BNE	R1	Loop	
<u>Register result status</u>												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
6		**72**	*Qi*	Load2		Mult1						

Loop Example Cycle 7

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1			Load1	Yes	80		
MUL	F4	F0	F2	1	2			Load2	Yes	72		
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2	6			Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	No			
SD	F4	0	R1	2				Store3	No			
<u>Reservation Stations</u>					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUBI	R1	R1	#8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1	Loop	
<u>Register result status</u>												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
7		**72**	*Qi*	Load2		Mult2						

Loop Example Cycle 7

	Instructio		j	k	iteration	Issue	complet	Result		Busy	Address		
In	LD	F0	0	R1	1	1			Load1	Yes	80		
LI	MUL	F4	F0	F2	1	2			Load2	Yes	72		
M	SD	F4	0	R1	1	3	4		Load3	No		Qi	
SI	SUB	R1	R1	#8	1	4	5	6					
LI	BNE	R1	Loop		1	5	7	-					
M	LD	F0	0	R1	2	6			Store1	Yes	80	Mult1	
SI	MUL	F4	F0	F2	2	7			Store2	No			
R	SD	F4	0	R1	2				Store3	No			
	Reservation Stations					S1	S2	RS for	RS for k				
		Time	Name	Busy	Op	Vj	Vk	Qj	Qk	Code:			
		0	Add1	No						LD	F0	0	R1
		0	Add2	No						MUL	F4	F0	F2
		0	Add3	No						SD	F4	0	R1
		0	Mult1	Yes	MULTD		R(F2)	Load1		SUB	R1	R1	#8
R		0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1	Loop	
C	Register result status												
	Clock		**R1**		F0	F2	F4	F6	F8	F10	F12	...	F30

Loop Example Cycle 8

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1			Load1	Yes	80		
MUL	F4	F0	F2	1	2			Load2	Yes	72		
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2	6			Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8			Store3	No			
Reservation Stations					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUBI	R1	R1	#8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
8		**72**	*Qi*	Load2		Mult2						

Loop Example Cycle 9

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1	9		Load1	Yes	80		
MUL	F4	F0	F2	1	2			Load2	Yes	72		
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2	6			Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load1		SUBI	R1	R1	#8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNE	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
9		**72**	*Qi*	Load2		Mult2						

Loop Example Cycle 10

Instructior		j	k	iteration	Issue	complet	Result		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2			Load2	Yes	72		
SD	F4	0	R1	1	3	4		Load3	No		Qi	
LD	F0	0	R1	2	6	10		Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					S1	S2	RS for	RS for k				
	Time	Name	Busy	Op	Vj	Vk	Qj	Qk	Code:			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD	M(80)	R(F2)			SUBI	R1	R1	#8
	0	Mult2	Yes	MULTD		R(F2)	Load2		BNEZ	R1	Loop	
Register result status												
Clock		**R1**		F0	F2	F4	F6	F8	F10	F12	...	F30
10		**72**	Qi	Load2		Mult2						

Loop Example Cycle 11

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2			Load2	No			
SD	F4	0	R1	1	3	4		Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					*S1*	*S2*	*RS for*	*RS for*	*k*			
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	4	Mult1	Yes	MULTD	M(80)	R(F2)			SUBI	R1	R1	#8
	0	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F1*	*...*	*F30*
11		**64**	*Qi*			Mult2						

Loop Example Cycle 12

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2			Load2	No			
SD	F4	0	R1	1	3	4		Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	3	Mult1	Yes	MULTD	M(80)	R(F2)			SUBI	R1	R1	#8
	4	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	*...*	*F30*
12		**64**	*Qi*			Mult2						

Loop Example Cycle 13

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2			Load2	No			
SD	F4	0	R1	1	3	4		Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	2	Mult1	Yes	MULTD	M(80)	R(F2)			SUB	R1	R1	#8
	3	Mult2	Yes	MULTD	M(72)	R(F2)			BNE	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
13		**64**	*Qi*			Mult2						

Loop Example Cycle 14

Instructior		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2	14		Load2	No			
SD	F4	0	R1	1	3	4		Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	Mult1	
MUL	F4	F0	F2	2	7			Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					*S1*	*S2*	*RS foi*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	1	Mult1	Yes	MULTD	M(80)	R(F2)			SUBI	R1	R1	#8
	2	Mult2	Yes	MULTD	M(72)	R(F2)			BNEZ	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
14		**64**	*Qi*			Mult2						

Loop Example Cycle 15

Instructior		j	k	iteration	Issue	complet	Result		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2	14	15	Load2	No			
SD	F4	0	R1	1	3	4		Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	M(80)*R	
MUL	F4	F0	F2	2	7	15		Store2	Yes	72	Mult2	
SD	F4	0	R1	2	8	9		Store3	No			
<u>Reservation Stations</u>					*S1*	*S2*	*RS for*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	No						SUBI	R1	R1	#8
	1	Mult2	Yes	MULTD		M(72)	R(F2)		BNE	R1	Loop	
<u>Register result status</u>												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F1*	*...*	*F30*
15		**64**	*Qi*			Mult2						

Loop Example Cycle 16

Instruction		*j*	*k*	*iteration*	*Issue*	*complet*	*Result*		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MULT	F4	F0	F2	1	2	14	15	Load2	No			
SD	F4	0	R1	1	3	4	16	Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	M(80)*R	
MULT	F4	F0	F2	2	7	15	16	Store2	Yes	72	M(72)*R	
SD	F4	0	R1	2	8	9		Store3	No			
Reservation Stations					*S1*	*S2*	*RS for j*	*RS for k*				
	Time	*Name*	*Busy*	*Op*	*Vj*	*Vk*	*Qj*	*Qk*	*Code:*			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MULT	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load3		SUBI	R1	R1	#8
	0	Mult2	No						BNEZ	R1	Loop	
Register result status												
Clock		**R1**		*F0*	*F2*	*F4*	*F6*	*F8*	*F10*	*F12*	...	*F30*
16		**64**	*Qi*			Mult1						

Loop Example Cycle 17

Instructior		j	k	iteration	Issue	complet	Result		Busy	Address		
LD	F0	0	R1	1	1	9	10	Load1	No			
MUL	F4	F0	F2	1	2	14	15	Load2	No			
SD	F4	0	R1	1	3	4	16	Load3	Yes	64	Qi	
LD	F0	0	R1	2	6	10	11	Store1	Yes	80	M(80)*R	
MUL	F4	F0	F2	2	7	15	16	Store2	Yes	72	M(72)*R	
SD	F4	0	R1	2	8	9	17	Store3	No			
Reservation Stations					S1	S2	RS for	RS for k				
	Time	Name	Busy	Op	Vj	Vk	Qj	Qk	Code:			
	0	Add1	No						LD	F0	0	R1
	0	Add2	No						MUL	F4	F0	F2
	0	Add3	No						SD	F4	0	R1
	0	Mult1	Yes	MULTD		R(F2)	Load3		SUBI	R1	R1	#8
	0	Mult2	No						BNEZ	R1	Loop	
Register result status												
Clock		R1		F0	F2	F4	F6	F8	F10	F12	...	F30
17		**64**	Qi			Mult1						